

MATH – 241
Engineering Mathematics I
Problem Set – 2

Question 1. Determine the values of λ such that the following linear equation system of three unknowns has only zero solution.

$$\begin{aligned}\lambda x_1 + x_2 + x_3 &= 0 \\ x_1 + \lambda x_2 + x_3 &= 0 \\ x_1 + x_2 + x_3 &= 0.\end{aligned}$$

Question 2. Determine the value of λ such that the following linear equation system of three unknowns has nonzero solutions.

$$\begin{aligned}x_1 + 2x_2 - 2x_3 &= 0 \\ 2x_1 - x_2 + \lambda x_3 &= 0 \\ 3x_1 + x_2 - x_3 &= 0.\end{aligned}$$

Question 3. Find the general solutions of the linear system of five unknowns.

$$\begin{aligned}x_1 + x_2 + x_5 &= 0 \\ x_1 + x_2 - x_3 &= 0 \\ x_3 + x_4 + x_5 &= 0.\end{aligned}$$

Question 4. Prove that if we add a new equation to a linear system of equations its solution set is a subset of the original system.

Question 5. Is it always possible that equations of inconsistent LSE (Linear System of Equations) can be separated into two sets of LSEs such that both are consistent ?

Question 6. Is it always possible that a consistent LSEs can be reduced to a unique solution case by eliminating some variables (i.e, equating them to zero) ?